

Research Interests

My research focuses on developing **human-like and trustworthy Artificial Intelligence** through the integration of computer vision, multimodal modeling, and data-driven problem solving. I am particularly interested in building **robust and interpretable vision-language systems** that can operate reliably in real-world environments, with emphasis on:

- (1) **Efficient and reliable visual understanding**, including object detection and lightweight architectures for real-time inference;
- (2) **Multimodal fusion** of visual, linguistic, and contextual signals for improved scene interpretation and decision-making;
- (3) **Behavioral and environmental analysis** from static and dynamic visual data to support real-world applications such as disaster assessment;
- (4) **Trustworthy AI**, ensuring fairness, robustness, privacy-preserving mechanisms (e.g., Differential Privacy), and safe deployability of machine learning and large vision-language models.

Education

Sogang University (Transferred 2024)

Seoul, South Korea

B.S. IN COMPUTER SCIENCE

Expected Aug. 2026

- Relevant Coursework: Deep Learning, Machine Learning, Big Data Computing

Hankuk University of Foreign Studies (HUFS)

Seoul, South Korea

INFORMATION & COMMUNICATION ENGINEERING

2020 – 2023

- Transferred to Sogang University in 2024

Projects

MULTI-TRACK VIDEO-TO-AUDIO DESCRIPTION (4 MEMBERS)

Mar 2026 – Present

- Developed a hierarchical video analysis pipeline using **Vision-Language Models (VLMs)** to generate **multi-track audio metadata**, focusing on dialogue, sound effects, music, and ambience.
- Designed a **reference-based evaluation framework** to maintain strict consistency between feature representation and semantic understanding in multimodal integration.
- **Submitted to KCC 2026** and currently under review.
GitHub: https://github.com/sogang-v2a/v2a_inspect

CHEST X-RAY MULTI-DISEASE DETECTION SYSTEM (CXr-CAD) (5 MEMBERS, TEAM LEADER)

Mar 2026 – Present

- Developed a **multi-label thoracic disease diagnosis system** using deep learning models, addressing class imbalance through **Focal Loss** and enhancing interpretability via **Grad-CAM**.
- Proposed a **deployment-readiness evaluation framework** focusing on model calibration, robustness, and resistance to data distribution shifts to ensure reliability in clinical environments.
- **Submitted to KCC 2026** and currently under review.
GitHub: <https://github.com/sogang-cxr-cad/CXR-CAD>

RAG-BASED, AI-POWERED CRM MANAGER (5 MEMBERS, TEAM LEADER)

Sep 2025 – Dec 2025

- Architected a **Multimodal RAG** pipeline on GCP, integrating GPT-4o (Vision/Whisper) for unified processing of text and image customer data with ChromaDB vector storage.
- Engineered a **self-evolving loop** using PostgreSQL logs and dynamic prompting, enabling the agent to learn from interactions and align with owner policies.
- Deployed **autonomous agent tools** (e.g., CS replies, BI reports) through a full-stack architecture, featuring a FastAPI backend and an interactive frontend dashboard for admins and agents.
GitHub: github.com/suhwantcha/CS-Agent

EFFICIENTNET DOG & CAT CLASSIFICATION (4 MEMBERS)

Dec 2024

- Implemented image classification on the Oxford-IIIT Pet Dataset using transfer learning with **EfficientNet-B0**, optimizing data augmentation and regularization (CutMix) and achieving **88.08% validation accuracy**.
GitHub: github.com/suhwantcha/efficientnet-Image-classification

YOLO OBJECT DETECTION (YOLOv5, YOLOv7, AND YOLOv11)

Mar 2025

- Trained and evaluated YOLO models (**YOLOv5, YOLOv7, and YOLOv11**) for object detection on the PASCAL VOC dataset, including data pre-processing (XML to .txt), transfer learning, and performance evaluation using mAP, optimized for real-time detection.
GitHub: github.com/suhwantcha/yolo-object-detection

- YOLO-PSYCHO-ANALYSIS** (PSYCHOLOGICAL ANALYSIS FROM HOUSE SKETCHES)

Apr 2025 – May 2025

 - Developed a **YOLOv8**-based model to analyze psychological characteristics by detecting elements such as doors, windows, sun, and chimney in user-drawn house sketches, converting them into quantitative scores and generating customized analysis scripts based on HTP test theory.
GitHub: github.com/suhwantcha/YOLO-Psycho-Analysis
- PINTOS OS**

Sep 2025 – Dec 2025

 - Implemented a **Pintos OS kernel** in C, including user program execution, file system support, and **priority-based thread scheduling**, with system calls, per-thread file descriptor management, and lock-based synchronization for safe concurrent file access.
GitHub: github.com/suhwantcha/OS-Pintos-Project

Experience

- Teaching Assistant, Basic Artificial Intelligence Programming**

SOGANG UNIVERSITY

Seoul, South Korea

Mar 2026 – Jun 2026

 - Supported Python programming coursework by handling Q&A, grading assignments, and managing attendance.
- Teaching Assistant, Understanding Artificial Intelligence**

SOGANG UNIVERSITY

Seoul, South Korea

Sep 2025 – Dec 2025

 - Supported an Introduction to Artificial Intelligence course for 50+ students, managing attendance, Q&A, and grading.
 - Conducted PyTorch-based labs on deep learning architectures, including CNNs and Transformers.
- Teaching Assistant, Computer Architecture**

SOGANG UNIVERSITY

Seoul, South Korea

Mar 2025 – Jun 2025

 - Supported course delivery for Computer Architecture and assisted 40+ students in understanding computer systems fundamentals.
- Member & Mentor, Parrot Data Science Club**

SOGANG UNIVERSITY

Seoul, South Korea

Sep 2024 – Present

 - Participated in data science and AI study sessions involving data analysis and deep learning projects.
 - Mentored 5 junior members (Mar 2025 – Jun 2025) in data analysis and CNN-based computer vision projects.
 - Organized internal Kaggle-style ML competitions and technical workshops.
- Teaching Assistant, English & Mathematics**

MEGASTUDY RUSSEL & YESUM ACADEMY

Seoul, South Korea

Feb 2024 – Present

 - Tutored 30+ students in analytical problem-solving and critical thinking.
 - Supported classroom management and lesson preparation for middle/high school students.
- Satellite Operation Squad Leader**

REPUBLIC OF KOREA ARMY

Daejeon, South Korea

Jan 2022 – Jul 2023

 - Ensured stable communication across military branches by operating the Military Satellite Communication System.
 - Maintained mission stability and rapid response in combat readiness operations, ensuring uninterrupted communication services.
 - Led a team of 7 members, overseeing maintenance, inspection, and operation of satellite equipment to enhance mission readiness.

Activities

- Autonomous Taxi Design Project**

SOGANG UNIVERSITY

Seoul, South Korea

Jul 2025 – Aug 2025

 - Designed an autonomous taxi system, focusing on AI-based navigation and safety protocols.
 - Developed cost-effective approaches for sensor fusion and environmental perception.
- Runner-up, Sogang AI Data Analysis Runnerthon**

SOGANG UNIVERSITY

Seoul, South Korea

Jan 2025 – Feb 2025

 - Built a real estate price forecasting model in a team of 4.
 - Achieved 2nd place among 15 teams using an XGBoost-based solution.
- Software and AI Competency Enhancement Training Program**

REPUBLIC OF KOREA ARMY

South Korea

Jan 2023 – May 2023

 - Completed intensive training in software development and AI applications.
 - Developed a weather forecasting model using real-world data and machine learning techniques.
- English Training Program (R-EIP)**

HANKUK UNIVERSITY OF FOREIGN STUDIES (HUFS)

Seoul, South Korea

Oct 2021 – Dec 2021

 - Completed an intensive academic English program focused on communication skills.

Technical Skills

Languages: Python, C/C++, SQL, R

ML/DL: PyTorch, TensorFlow, scikit-learn, Hugging Face, LangChain

Data & Vision: NumPy, Pandas, OpenCV, Pillow, Seaborn

Tools: Git, Linux, VS Code, Jupyter Notebook, Google Colab, MySQL, CLion, Notion

Languages

Korean: Native

English: Fluent (TOEFL iBT 100, Feb 2026)

Japanese: Basic (Studying)